

Выпуклые функции: субдифференциал и двойственность

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A convex function is a convex set — its epigraph — so the whole geometry of the previous sheet applies here as well. The one new tool is the Legendre–Fenchel transform: it turns a function into the family of its supporting lines and makes duality visible.

Definition 1. $f: \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is **convex** if $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$ for all x, y and $\lambda \in (0, 1)$; equivalently, its epigraph $\{(x, t) : t \geq f(x)\}$ is a convex subset of $\mathbb{R}^{n+1}$.

Definition 2. The **subdifferential** of f at x is $\partial f(x) = \{p \in \mathbb{R}^n : f(y) \geq f(x) + \langle p, y-x \rangle \ \forall y\}$; its elements are the slopes of supporting affine functions at x .

Definition 3. The **Legendre–Fenchel transform** of f is $f^*(p) = \sup_x (\langle p, x \rangle - f(x))$.

Theorem 1 (Jensen 1906). If f is convex and μ is a probability measure with barycentre $\bar{x} = \int x d\mu$, then $f(\bar{x}) \leq \int f d\mu$.

Theorem 2 (Fenchel–Moreau). If f is convex, lower semicontinuous and not identically $+\infty$, then $f^{**} = f$. In particular $\langle p, x \rangle \leq f(x) + f^*(p)$ for all x, p (Young’s inequality), with equality exactly when $p \in \partial f(x)$.

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1. Let f be convex and finite on an open interval $I \subseteq \mathbb{R}$. Prove that the one-sided derivatives $f'_-(x) \leq f'_+(x)$ exist everywhere, are nondecreasing, and that f is locally Lipschitz on I . Deduce that a finite convex function

on an open set of $\mathbb{R}^n$ is continuous, and give an example showing that this fails on a non-open domain.

2. (Jensen 1906) Prove Theorem 1 for finitely supported measures and deduce the weighted AM–GM inequality. Then prove the general case, and show that for strictly convex f equality forces μ to be a point mass.
3. (Proposed problems, №314) Let $f: [0, 1] \rightarrow \mathbb{R}$ be convex and of class C^2 . Prove that

$$\int_0^1 f(x) dx \leq \frac{1}{4} \left(f(0) + f\left(\frac{1}{3}\right) + f\left(\frac{2}{3}\right) + f(1) \right).$$

4. (MIPT 2010) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable and suppose that $f(x) - \varepsilon \|x\|^2$ is convex for some $\varepsilon > 0$. Let x_0 be a point of global minimum of f . Prove that $f(x) - f(x_0) \geq \varepsilon \|x - x_0\|^2$ for all x .
5. (Fenchel 1949) Prove Young's inequality $\langle p, x \rangle \leq f(x) + f^*(p)$ directly from the definition, and compute f^* for

$$f(x) = \frac{|x|^p}{p} \ (p > 1), \quad f(x) = e^x, \quad f = \mathbf{1}_K^\infty \text{ (the convex indicator of a convex set } K\text{)}.$$

Which classical inequality does the first computation reproduce?

6. (Капачёв 1.76) Let $(a_n)_{n \geq 0}$ be a positive decreasing sequence tending to zero which is convex, i.e. $a_{n-1} - 2a_n + a_{n+1} \geq 0$ for $n \geq 1$. Prove that $na_n \rightarrow 0$.